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Challenges of Gate-Based Quantum Optimization for a Discrete Process-Design Problem

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ABSTRACT

Discrete decisions, such as selecting routes, units, and operating modes, are an early and combinatorially hard step in process design. Gate-based quantum optimization algorithms, namely the quantum approximate optimization algorithm (QAOA) and the variational quantum eigensolver (VQE) are increasingly proposed for such problems, raising a practical question for the process systems engineering community. Once a discrete-design model is mapped to the quadratic unconstrained binary optimization (QUBO) form required by these methods, do the resulting samples actually correspond to process-design decisions, and how do they compare with classical methods? We study this question on a small but fully enumerable instance of drug-substance manufacturing with 19 binary flow decisions. The penalty reformulation needed to obtain a QUBO adds 17 auxiliary variables, so a sampled 36-bit state is not itself a design decision: it becomes one only after projection onto the 19 flow variables and exact repair of the auxiliaries. After this preprocessing, energy-targeted transferred-angle QAOA concentrates probability on the optimal flowsheet (664 of 262144 ideal reads), but does not outperform a simple classical hill climber, an exact solver, or exhaustive enumeration. A small IBM hardware run completes the same pipeline without adding an advantage. We report these results as a workflow-feasibility and interpretation study, not as a claim of quantum speedup.

Keywords: quantum computing, process design, discrete optimization, QUBO reformulation, pharmaceutical manufacturing

INTRODUCTION

Discrete choices are often the first barrier in process design. Before a flowsheet can be simulated or optimized over continuous variables, the designer must commit to a set of mutually exclusive options: which synthesis route to use, which unit operations to include, and which operating modes to run [1]. In pharmaceutical manufacturing, these choices are particularly consequential because route and equipment selection shape cost, yield, and regulatory burden, and because digital design and optimization tools for this sector are comparatively recent [2, 3]. The resulting models are combinatorial, and even modestly sized superstructures generate large discrete search spaces.

Gate-based quantum optimization is one of the emerging tools proposed for this setting. A gate-based quantum computer operates on quantum bits (qubits) by applying sequences of parameterized gate operations that form a quantum circuit; measuring the circuit returns a binary string, and the circuit parameters, called angles, are tuned so that low-energy strings appear with high probability. The quantum approximate optimization algorithm (QAOA) [4] and the variational quantum eigensolver (VQE) [5, 6] follow this variational paradigm: QAOA uses p alternating cost and mixer layers, and VQE uses a parameterized circuit with structure similar to the quantum device, also known as a hardware-efficient ansatz; in both cases, a classical optimizer adjusts the angles. Current hardware is noisy and of intermediate scale [7], and benchmarking surveys stress that a practical quantum advantage for optimization has not yet been established [8]; the relevant question is what is required to use these methods today and how their output compares with classical baselines.

Within process systems engineering, quantum computing has been reviewed as an emerging capability [9], and hybrid quantum-classical strategies have been proposed for discrete optimization in process and energy systems [10]. Discrete-design instances have been mapped to the QUBO form [11] and solved with Ising-based annealing hardware [10, 12]. These works establish that the mapping is possible; few examine what happens to the samples after the reformulation, and the gate-based path is less explored than annealing. Casting a constrained problem as a QUBO introduces auxiliary variables [13] that are bookkeeping rather than design decisions, so a raw quantum sample is not directly interpretable as a flowsheet. The question is: what does it take to turn gate-based quantum samples into process-design decisions, and how does this compare with classical methods?

We answer this through an end-to-end audit on a single drug-substance manufacturing (DS-MFG) instance, chosen to be small enough to enumerate exactly yet large enough to expose the reformulation and interpretation issues that arise at scale. Our contribution is not a new QAOA or VQE variant. We find: (i) raw QUBO samples require projection and exact auxiliary repair before they are interpretable as design candidates; (ii) a quadratic surrogate over the design variables fits the repaired landscape globally but fails as an optimizer; (iii) transferred-angle QAOA concentrates probability on good flowsheets but does not beat classical baselines; and (iv) a hardware run confirms pipeline feasibility without adding advantage.

For the process systems engineering community, the value of this study is a concrete, reproducible measurement of the engineering burden that sits between a discrete-design model and a usable quantum sample, on a problem from a relevant application domain. The conclusion is deliberately conservative: the results demonstrate feasibility and interpretation, not a runtime or hardware advantage over classical computation.

METHODS

The Discrete Design Instance

The DS-MFG instance is the discrete subproblem of a drug-substance manufacturing case study introduced by Park and Bernal Neira [12], where a mixed-integer nonlinear design problem is decomposed so that the discrete part is handled by combinatorial solvers and the nonlinear part by simulators. We keep the same discrete instance but ask what is required to run gate-based optimizers. The instance is a binary integer program for capital-expenditure selection in the manufacturing superstructure: the 19 binary variables encode candidate material-flow arcs and unit-operation selections, and the objective function is a normalized capital-cost score. The discrete problem has the form

$$\begin{aligned}
\min_{f} \quad & \sum_{i\in F} c_i f_i \\
\text{s.t.} \quad & \sum_{i\in \mathrm{In}(n)} f_i = \sum_{j\in \mathrm{Out}(n)} f_j \qquad \forall n \in N, \\
& f_{\text{source}} = 1, \quad f_{\text{sink}} = 1, \\
& g(f) \le 0, \\
& f_i \in \{0,1\} \qquad \forall i \in F.
\end{aligned} \tag{1}$$

Equation (1) lists the relevant constraint classes: flow balance at each node $n \in N$, fixed source and sink selections, and additional logical constraints $g(f) \le 0$ that enforce a valid flowsheet.

Solved with `Gurobi` [14], the model (19 binary variables, 20 constraints) returns an optimal capital-cost score of 11.7095; with feasible solutions enumeration enabled (`PoolSearchMode=2`), all 36 feasible flows are retained in a single deterministic solve (0.073 s). We treat 11.7095 as the certified global optimum and use it as the reference throughout.

From Integer Program to QUBO and Back

To run QAOA and VQE, the integer program must be expressed as a QUBO, in which all constraints are folded into a single quadratic objective over binary variables [11, 13]. Inequality constraints require auxiliary (slack) variables; the penalty weights and auxiliary count are not unique, and both affect how faithfully the QUBO represents the original problem [13]. For this instance, the penalty reformulation introduces 17 auxiliary variables, so the QUBO acts on 36 binary variables. For a bitstring $x \in \{0,1\}^{36}$, the encoded objective is

$$E_{\text{QUBO}}(x) = \ell^{\top} x + x^{\top} Q x + b, \tag{2}$$

with linear coefficients ℓ, quadratic coefficient matrix Q, and constant offset b.

The 36 variables are separated into the original flow variables and the auxiliaries,

$$x = (y, z), \qquad y \in \{0,1\}^{19}, \quad z \in \{0,1\}^{17}.$$

Only y encodes a design decision; z is bookkeeping. A sampled 36-bit string does not map directly to a flowsheet. We make it interpretable by projecting onto y and repairing the auxiliaries, choosing the assignment that minimizes the QUBO energy for that flow,

$$F(y) = \min_{z\in\{0,1\}^{17}} E_{\text{QUBO}}(y,z). \tag{3}$$

The repair is exact and cheap here because, once y is fixed, the auxiliary-auxiliary coupling graph splits into 13 independent connected components: 9 singletons (one auxiliary bit each) and 4 pairs (two auxiliary bits each). The singletons correspond to the at-most-one inequality constraints for each decision group and to the downstream flow-balance conditions; the pairs arise from product-linearization inequalities for the Route B activation rule ($f_{06} = 1$ iff Reactor 1 is continuous and Reactor 2 is batch). Each component is solved by exhaustive enumeration over its 2 or 4 possible assignments; the $9\times2 + 4\times4 = 34$ total comparisons replace $2^{17} = 131{,}072$ brute-force evaluations. The full component table with auxiliary indices, flow neighbors, and constraint roles appears in the Supplement (Sect. 3). For a general QUBO, the auxiliary repair is itself NP-hard, a limitation we return to in the Discussion.

The repaired minimizer over all 2^{19} flows recovers the optimum, and the repaired QUBO scores match the integer-program objective to within 3.1×10^{-13}, confirming they encode the same objective. Exact enumeration finds exactly 36 feasible flows, as does the solution pool option of `Gurobi`. Note that 14 of the 50 lowest-cost repaired flows are infeasible flows whose penalty values are insufficient to place them above all feasible ones.

The repaired objective $F(y)$ is not a quadratic function of y, but QAOA and VQE require a QUBO input. Rather than using the full 36-variable QUBO (which mixes design variables with auxiliaries and requires 36-qubit circuits), we construct a reduced, 19-variable QUBO directly over the design space y by fitting a quadratic surrogate to the repaired landscape:

$$\widehat{F}(y) = \hat{b} + \hat{\ell}^{\top} y + y^{\top}\widehat{Q}y \tag{4}$$

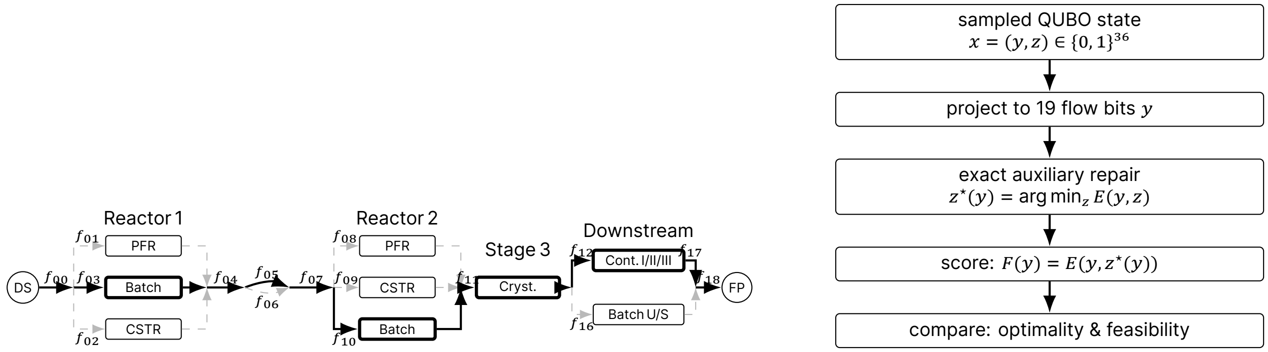
by least squares over all 2^{19} repaired flows. The surrogate fits the landscape almost perfectly in aggregate ($R^2 \approx 0.999$), but this global accuracy is misleading for optimization: the surrogate's own minimizer repairs to 14.5505 rather than the true 11.7095, the true optimum ranks only 33rd under the surrogate, and among the 50 lowest-cost repaired flows the surrogate's absolute errors reach 29.4. We therefore use $\widehat{F}$ only to generate samples via the quantum algorithms; every reported sample-quality metric is computed with the exact repaired objective F, not $\widehat{F}$.

Quantum, Classical, and Hardware Workflows

We evaluate several workflows over this pipeline. The local QAOA and VQE workflows run through `QiskitOpt.jl`, a Julia wrapper for Qiskit [15], with QUBO models constructed via the `QUBO.jl` ecosystem [13]. In QAOA, we varied the parameter of repetition of mixer and driving Hamiltonians, p, which, as it tends to infinity, converges to the adiabatic quantum algorithm [16] with convergence guarantees yet yields deeper quantum circuits, more prone to error in current NISQ devices [7]. Both algorithms have a number of other parameters, also called angles, that are found variationally by an external classical optimizer [6], in our case COBYLA. Direct full-QUBO QAOA and VQE operate on the 36-variable formulation; reduced-surrogate QAOA and VQE operate on the 19-variable surrogate and are scored by exact repair. The best-performing quantum workflow adds an offline parameter-search step: QAOA parameters are optimized on the surrogate with JuliQAOA [17] by minimizing the expected surrogate energy $\mathbb{E}_{y\sim p_\theta}[\widehat{F}(y)]$, then transferred to a Qiskit circuit for sampling (energy-targeted, non-oracle). The fixed, pre-optimized parameters with the highest local solution concentration are also submitted to the hardware for a feasibility test. Reduced-surrogate VQE uses an `EfficientSU2` ansatz optimized with COBYLA.

As classical baselines, we sample the repaired 19-flow objective by uniform random sampling and by single-bit-flip hill climbing with random restarts. As a hardware feasibility test, we run the highest-concentration fixed-parameter QAOA circuit on the IBM `ibm_fez` quantum computer. An independent state-vector calculation predicts the central QAOA reference-optimum probability at 2.63×10^{-3}, consistent with the 2.53×10^{-3} sampled rate. Software versions, random seeds, simulator settings, circuit depths, and job identifiers are recorded in the artifact bundle.

For each sampled state $x = (y,z)$ we record the raw QUBO energy $E_{\text{QUBO}}(y,z)$, the repaired objective $F(y)$, and the auxiliary gap. A *reference-optimum hit* means the repaired flow equals the certified optimal flowsheet (the unique minimum-cost feasible design). A *feasible-design hit* means the repaired flow matches any of the 36 feasible flows. We report both hit rates with two-sided Wilson 95% confidence intervals (score-based intervals for proportions that remain valid near zero); rows with no observed hits report the one-sided upper bound. The empirical 99% time-to-solution $\text{TTS}_{99} = \tau\lceil\log(0.01)/\log(1-\hat{p})\rceil$ normalizes each hit rate by the measured wall time per sample τ, enabling wall-clock comparison across methods with different per-sample costs; rows with zero hits report ∞.



(a) DS-MFG manufacturing superstructure (19 binary arcs)

(b) Repair-aware scoring pipeline

Figure 1. Overview of the DS-MFG case study. **(a)** Manufacturing superstructure: thick arrows trace the optimal flowsheet (Batch Rxn 1 → Route A (f_{05}) → Batch Rxn 2 → Cryst. → Cont. I (f_{13})); dashed arrows mark the non-optimal alternatives; the backbone arcs $f_{00}, f_{04}, f_{07}, f_{11}, f_{18}$ carry every feasible flow. Route B (f_{06}) is activated only when a continuous Reactor 1 feeds a batch Reactor 2; Cont. I/II/III share entry arc f_{12} and exit arc f_{17} (optimal uses f_{13}). **(b)** Every sampled 36-bit QUBO string is projected to the 19 flow variables, repaired exactly via the 13-component auxiliary decomposition (Supplement, Sect. 3), and scored against the certified optimum and the 36-member feasible-design pool; hit rates appear in Table 1.

RESULTS

Repair Changes the Meaning of a Sample

Sampling the 36-variable QUBO directly does not yield an interpretable design result. A $p = 2$ full-QUBO QAOA found no optimum in 512 samples; a 32768-read run returned it only 4 times after repair, and never as a fully optimal 36-bit string without projection. Without projection and exact repair, a raw QUBO sample cannot be read as a flowsheet; Figure 1(b) summarizes the scoring pipeline, and the repair decomposition is detailed in the Supplement (Sect. 3).

Reduction and Parameter Transfer Concentrate QAOA Samples

Once the pipeline is in place, reduction to the surrogate and transferred-parameter search are what make QAOA samples concentrate on feasible, optimal flowsheets. The energy-targeted $p = 5$ transferred workflow sampled the certified optimum 664 times in 262144 ideal reads and matched a feasible flowsheet 32502 times (12.4%, CI [12.3, 12.5]%), the highest feasible-design rate of any workflow; see Table 1 for TTS99 values.

VQE and Classical Baselines Bound the Claim

VQE benefits from the same reduction but produces a flatter distribution. Full-QUBO VQE found no optimum in a five-seed sweep; a selected-seed reduced-surrogate VQE found the optimum 20 times in 524288 reads. Hill climbing found the optimum 19 times in 262144 evaluations; despite a lower hit rate than QAOA, its much smaller per-sample cost gives a better TTS99-opt (0.23 s vs. 0.37 s). Random sampling never found the optimum.

Hardware Execution Is an Implementation Feasibility Check

The highest-concentration fixed-angle QAOA circuit ran on the `ibm_fez` quantum computer across three repeats and three random transpile seeds (92001-92003), with 4096 measurement shots per repeat, for 36864 total hardware reads (9 jobs, 73.5 s elapsed including cloud queue). The run produced 4 feasible-design reads and no reference-optimum read; the best hardware sample was the second-cheapest feasible flowsheet (objective 11.8105). The resulting TTS99-feas is 84.6 s, roughly 14000× slower than the hill climber's 0.006 s and 1200× slower than Gurobi's 0.073 s, primarily because most elapsed time is cloud overhead rather than circuit execution (≈2 ms per shot). With zero optimum hits, the Wilson 95% upper bound on the hardware optimum-hit rate is about 1.04×10^{-4}. Because calibration and full transpilation metadata were not retained, this run confirms only that the submission-and-scoring path executes end to end on hardware.

DISCUSSION AND LIMITATIONS

This study answers a narrower question: not whether quantum optimization solves the DS-MFG instance (Gurobi does in under a second), but what must happen for a QAOA or VQE distribution to yield meaningful process-design samples. That engineering path requires

Table 1: Primary outcomes, all scored after exact auxiliary repair. Opt rate: empirical probability of sampling the certified optimal flowsheet; feas rate: any feasible-design match among the 36 feasible flows. Wilson 95% CIs are shown as [lo, hi] in the same units as the point estimate; zero-hit rows report only the one-sided upper bound. Total time: aggregate wall time for all samples (Aer simulation only for local quantum methods, excludes offline preprocessing; total elapsed including cloud job queue for IBM hardware). TTS99-opt and TTS99-feas are empirical 99% time-to-solution (see Methods); ∞ if no hits observed.

Workflow	Total time (s)	Opt rate [95% CI]	Feas rate [95% CI]	TTS99-opt (s)	TTS99-feas (s)
Gurobi solution pool	0.073	Deterministic	36/36 (exhaustive)	0.073	0.073
Uniform random sampling	2.08	$<1.5\times10^{-5}$	$8.0\ [5.2,\ 12.2]\times10^{-5}$	∞	0.46
Hill-climb restarts	0.93	$7.2\ [4.6,\ 11.3]\times10^{-5}$	$2.75\ [2.56,\ 2.96]\times10^{-3}$	0.23	0.006
Energy-targeted QAOA, p=5	53.9	$2.53\ [2.35,\ 2.73]\times10^{-3}$	$0.124\ [0.123,\ 0.125]$	0.37	0.007
Reduced-surrogate VQE, seed 74018	30.1	$3.8\ [2.5,\ 5.9]\times10^{-5}$	$7.4\ [6.7,\ 8.2]\times10^{-4}$	6.93	0.36
IBM `ibm_fez` pilot	73.5	$<1.1\times10^{-4}$	$1.1\ [0.42,\ 2.8]\times10^{-4}$	∞	84.6

variable separation, exact auxiliary repair, a surrogate for the QUBO interface, offline angle search, and scoring in the original flow space. The repaired distribution does concentrate on good flowsheets; it does not translate into a runtime advantage over classical methods for this instance.

The main limitations follow directly. Exact auxiliary repair is tractable here only because the conditioned auxiliary graph decomposes into 13 independent components (Supplement, Sect. 3); in general, it is NP-hard, and an approximate repair would change the meaning of every reported metric. Read counts exclude the offline preprocessing that dominates the real cost, and meaningful comparison with classical optimization requires accounting for that total cost [8, 18].

CONCLUSION

For this drug-substance manufacturing case study, QAOA and VQE samples become interpretable as process-design decisions only after the QUBO auxiliary variables are separated from the flow variables and repaired exactly. With that bridge in place, energy-targeted transferred-angle QAOA concentrated probability on the optimal flowsheet (664 of 262144 ideal reads) and returned a feasible flowsheet 32502 times, while a selected-seed VQE follow-up reached the optimum 20 times in 524288 reads. A small IBM hardware run completed the same pipeline and found a feasible but non-optimal flowsheet. None of these outperforms a simple classical hill climber or the exact solver.

The lesson for the process systems engineering community is that the engineering effort between a discrete-design model and a usable quantum sample is substantial and easy to understate, and that honest evaluations of gate-based quantum optimization for process design require scoring in the original decision space and reporting the full preprocessing cost.

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DECLARATION OF USE OF AI

The authors used generative AI tools to assist in generating initial code, data analysis workflows, and draft visualizations as described in the relevant sections of the manuscript. All AI-generated outputs were reviewed, tested, and modified by the authors. The authors take full responsibility for the accuracy, validity, and integrity of all results.

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